# CS 185/285 HW1 Report: Imitation Learning on Push-T

All runs use the default `TrainConfig`: chunk size K = 8, batch size 128, Adam (lr 3e-4, no weight decay), 400 epochs (75,600 gradient steps), seed 42. Each policy is evaluated every 10,000 steps and once more at the end of training. Each evaluation is 100 episodes, and the reported metric is the mean over episodes of each episode's maximum reward. Flow matching uses 10 Euler integration steps at inference.

| Policy | Final eval reward (step 75,600) | Best eval reward | Target |
|---|---|---|---|
| MSE | **0.595** | 0.641 (step 60k) | ≥ 0.5 |
| Flow matching | **0.845** | 0.868 (step 70k) | ≥ 0.7 |

## 1. Action chunking with MSE loss

### Architecture

The MSE policy is an MLP that maps the 5-D normalized state to the whole normalized action chunk in one forward pass:

```
Linear(5 → 256) → ReLU → Linear(256 → 256) → ReLU → Linear(256 → 256) → ReLU →
Linear(256 → 16)
```

- 3 hidden layers of width 256 with ReLU activations, and a linear output layer.
- The 16 outputs are reshaped to a (K = 8) × (action dim = 2) chunk. There are 137,232 parameters.
- Loss (Eq. 1): the squared L2 error summed over the whole chunk, averaged over the batch.
- States and actions are z-score normalized with dataset statistics. Predicted actions are de-normalized and clipped to the action space before execution.

### Training curves

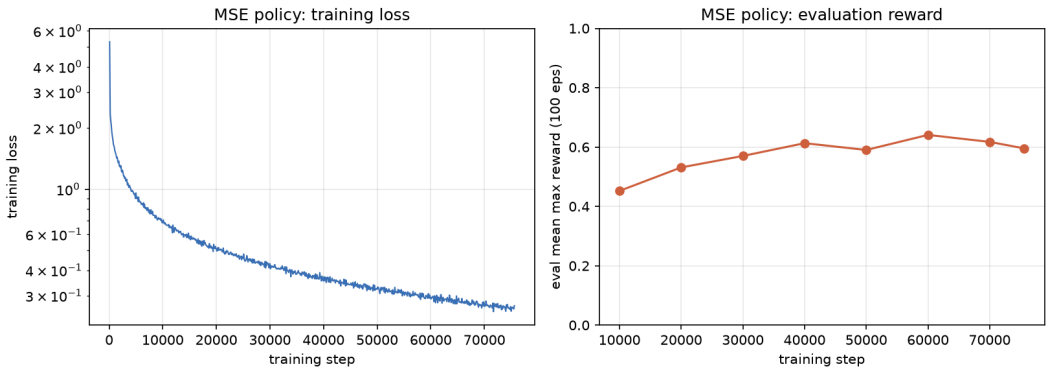



Figure 1: MSE training curves

The training loss falls from about 5.3 to 0.27. The eval reward passes 0.5 by step 20k and then plateaus around 0.6 (between 0.59 and 0.64), even though the training loss keeps decreasing.

## 2. Action chunking with flow matching

### Architecture

The flow matching policy uses the same MLP body: 3 × 256 hidden units, ReLU, and 141,584 parameters. The input is the concatenation of three things: the state (5), the flattened noisy chunk $A_\tau$ (16), and the flow timestep $\tau$ (1), for 22 inputs in total. The output is the 16-D velocity $v_\theta(o, A_\tau, \tau)$.

- **Training (Eq. 2):** sample $A_0 \sim N(0, I)$ and $\tau \sim U(0, 1)$. Set $A_\tau = \tau A + (1 - \tau)A_0$ and regress $v_\theta$ onto the target $A - A_0$ with a summed squared error.
- **Inference (Eq. 3):** start from $A_0 \sim N(0, I)$ and take $n = 10$ Euler steps, $A \leftarrow A + (1/n) \cdot v_\theta(o, A, i/n)$ for $i = 0, \ldots, n - 1$.

### Training curves

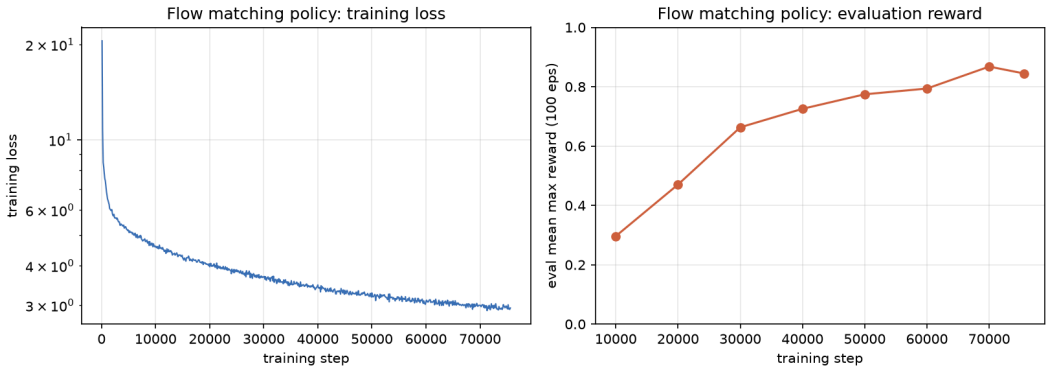



Figure 2: Flow matching training curves

The flow-matching loss stays higher than the MSE loss (about 2.9 at the end). This is expected: the regression target $A - A_0$ contains random noise, so the loss cannot reach zero. The eval reward climbs steadily, from 0.30 at step 10k, past 0.7 at step 40k, to 0.87 at step 70k.

## 3. Qualitative comparison (from the rollout videos)

These observations come from the five recorded rollouts at the final checkpoint. Both policies start from the same initial states (eval seeds 0–4).

- **The MSE policy** quickly pushes the T roughly into the goal region. Once the T is nearly aligned, the agent often **stalls**: it hovers next to the T, or sits on a corner, and stops making progress. In episodes like this it stays nearly still for the rest of the 300-step time limit. This fits the averaging behavior of an MSE regressor. When the demonstrations contain several valid ways to continue (for example, pushing a corner from the left or from the right), the L2-optimal prediction is their mean. That mean is a small, indecisive motion that does not resolve the remaining position or orientation error.
- **The flow matching policy** commits to one mode at a time. The agent keeps moving around the T, repositioning to different contact points and making repeated small corrective pushes. The block ends up more precisely overlapped with the green target, which is why its maximum reward is clearly higher. Because each chunk is sampled from a learned multimodal distribution, consecutive chunks can choose different strategies. Its motion therefore looks less smooth and more exploratory than the MSE policy's, but it rarely freezes. Neither policy usually reaches the environment's strict success threshold before the time limit: 4 of the 5 flow videos and 3 of the 5 MSE videos run the full 300 steps. The difference is that the flow policy keeps improving the overlap instead of stalling.

Overall, the extra expressiveness of flow matching, which can model a multimodal action-chunk distribution, translates directly into higher task success on Push-T: 0.845 vs. 0.595 final reward.